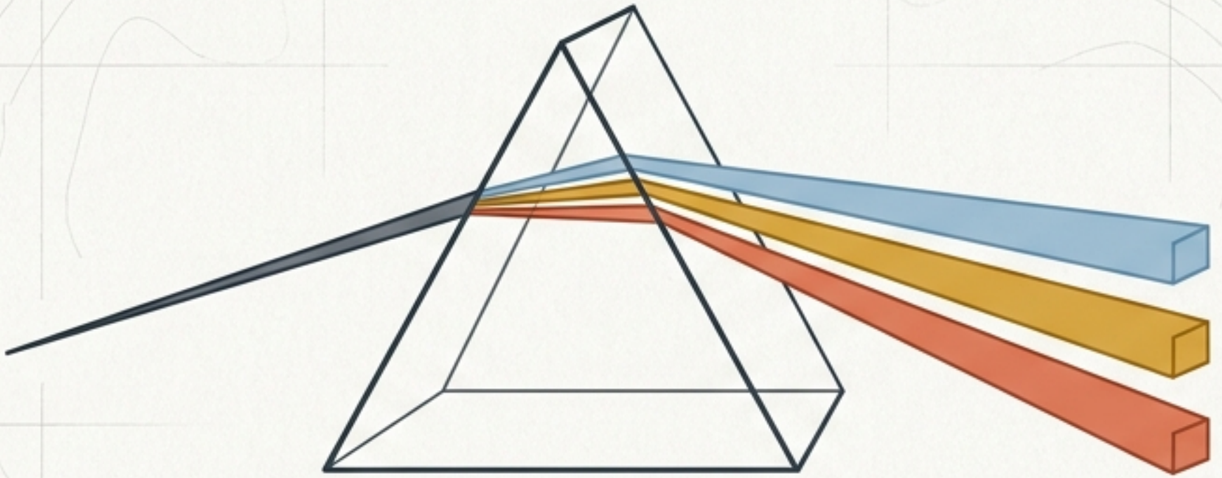


Evaluation as Evidence

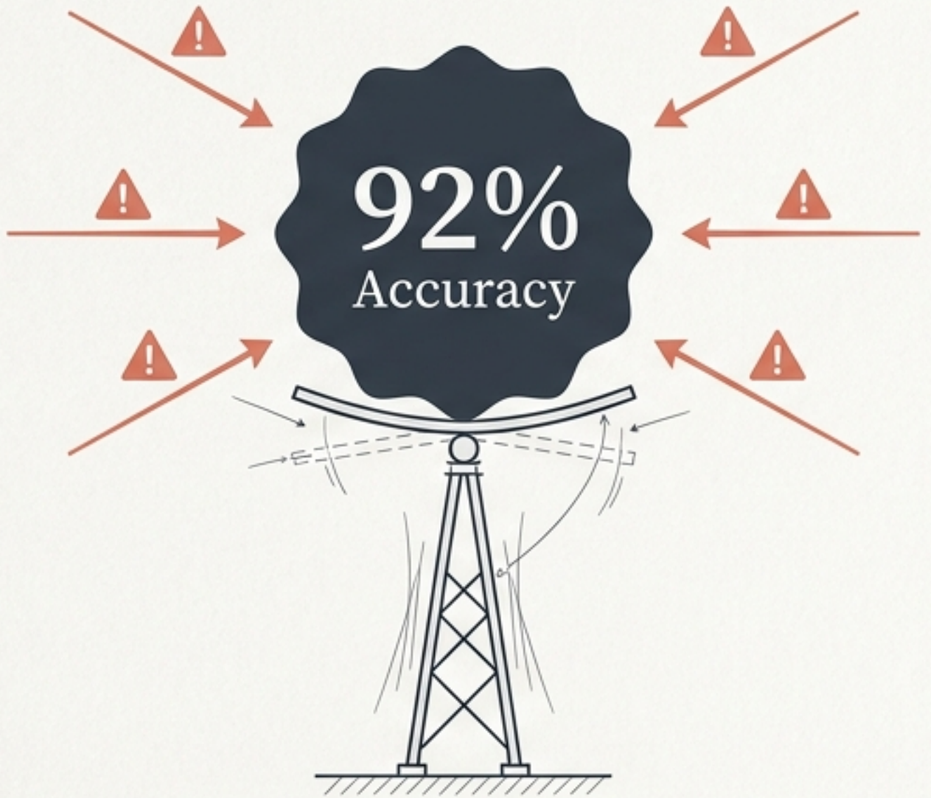
Generalization, Robustness, and the Anatomy of Distribution Shift.

A manifesto for responsible machine learning deployment.

Gemini Notebook

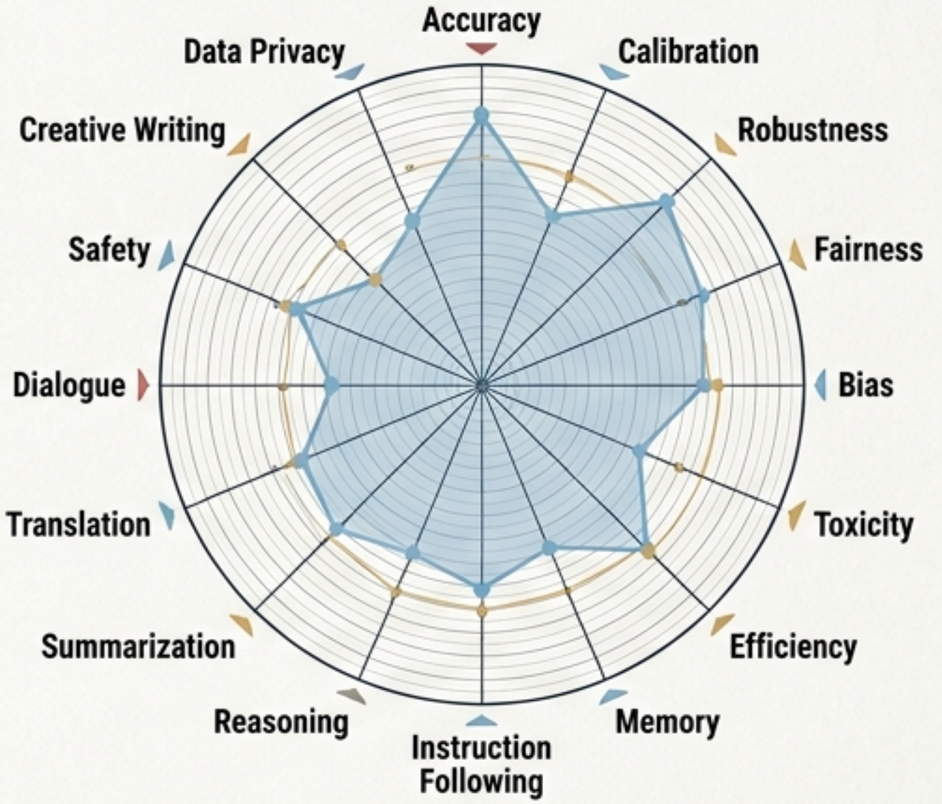
The myth of the single test score

One-Dimensional Instability



Pretraining exposes foundation models to broad, incompletely documented data. Prompting creates infinite downstream systems. A single average score cannot characterize this massive behavioral surface.

Multi-Metric Robustness (Stanford HELM)



Stanford's HELM project explicitly replaced the single score with a multi-metric evaluation across 16 scenarios, treating incomplete coverage as a fact to document, not a weakness to conceal.

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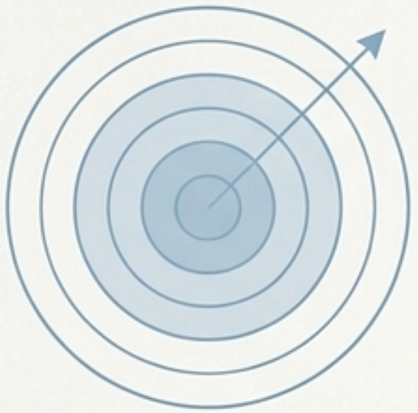
Disambiguating the core triad



Generalization

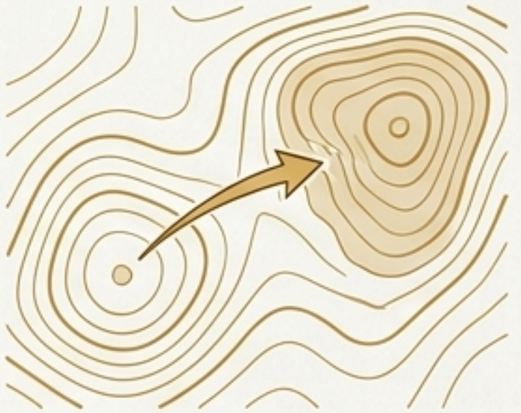
Focuses on performance beyond the training sample.

Can the model handle new data generated by the exact same procedure?

Robustness

Focuses on acceptable performance under specified perturbations.

Does the model break when specific, defined elements of the problem change?



Distribution Shift

Focuses on a probability mismatch between training and deployment.

What happens when the fundamental probability distributions generating the data change?

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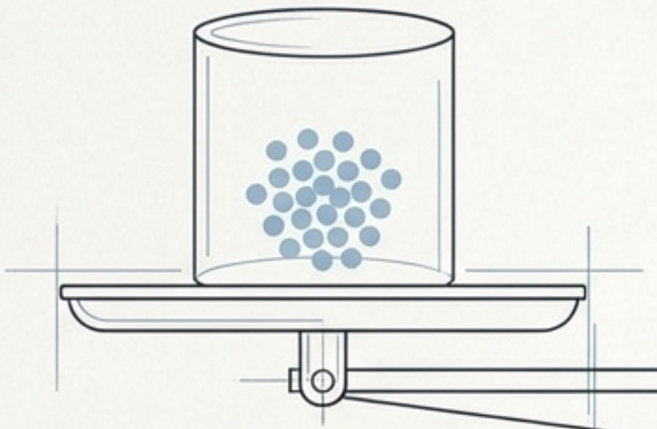
The gap between expectation and reality

Empirical Risk



$$\hat{R}_{\text{train}}(f_\theta) = \frac{1}{n}\sum \ell(f_\theta(x_i), y_i)$$

Finite observations.
Classical IID assumption.

Population Risk

$$R_P(f_\theta) = \mathbb{E}_{(X,Y)\sim P}\left[\ell(f_\theta(X), Y)\right]$$

The true, unobservable population distribution.

Core insight: A held-out test set only estimates population risk. Finite samples produce sampling error. More test examples reduce noise, but they cannot repair an unrepresentative test distribution.

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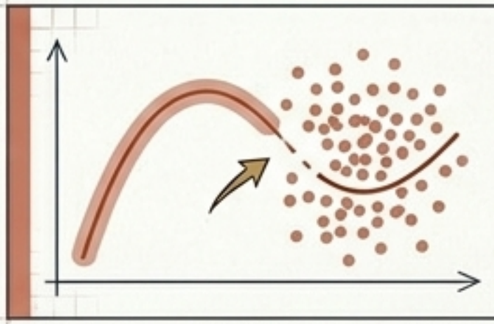
Three mechanical sources of failure

Model Failure

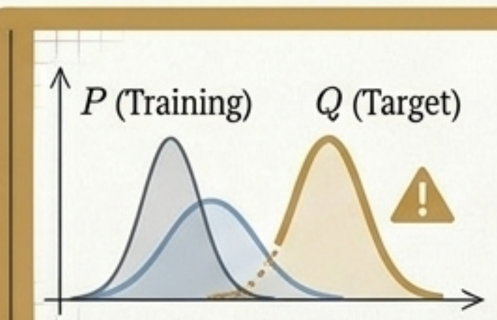
1. Estimation Error

The test sample is simply too small or too noisy to estimate performance precisely.



2. Optimization Error

Training fails to find a suitable predictor, or the chosen model family cannot represent the true underlying relationship.



3. Distribution Mismatch

The most dangerous failure. The relevant deployment risk is $R_Q(f)$ for a target distribution Q, but evaluation only estimated $R_P(f)$. Evaluation measured the wrong world.

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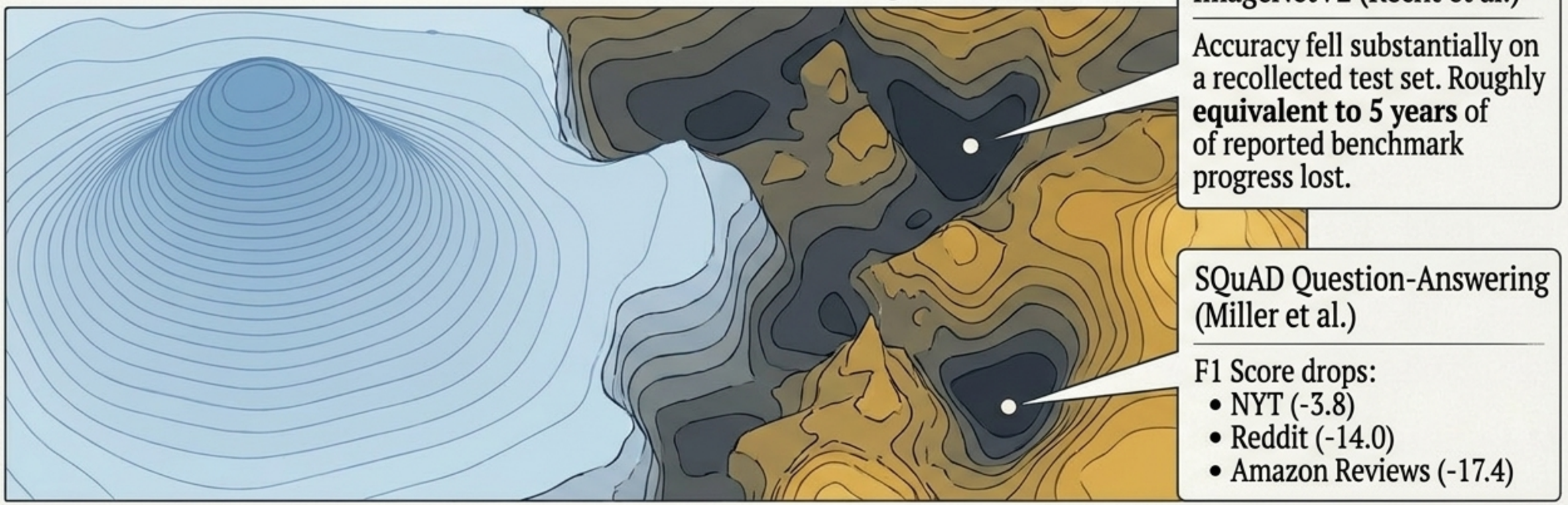
Generalization is strictly relative to a distribution



ImageNetV2 (Recht et al.)

Accuracy fell substantially on a recollected test set. Roughly **equivalent to 5 years** of of reported benchmark progress lost.

SQuAD Question-Answering (Miller et al.)

F1 Score drops:

- NYT (-3.8)
- Reddit (-14.0)
- Amazon Reviews (-17.4)

Takeaway: A small train-test gap only proves a model behaves similarly on two samples constructed by the same procedure. The SQuAD study separates two often-conflated hypotheses: benchmark overuse versus fundamental domain change exposing non-generalized features.

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The underspecification problem

Feature Set A: Stable relationships

Relies on genuine, causally linked mechanisms.

Feature Set B: Background shortcuts

Exploits visual watermarks, medical device artifacts, or documentation style.

Feature Set C: Demographic proxies

Leverages correlations with nuisance attributes.

Many predictors achieve almost identical validation loss while relying on totally different features. Successful optimization does not uniquely determine a reliable solution. An empirical study (D'Amour et al.) spanning multiple domains found that models with identical in-domain performance differed materially under real-world stress tests.

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A practical taxonomy of distribution shift

Shift Type	What Changes? (Math)	Assumed Invariant	Real-World Scenario
Covariate	$P_{\text{train}}(\mathbf{X}) \neq P_{\text{test}}(\mathbf{X})$	$P(Y\|\mathbf{X})$	A new clinic serves different proportions of otherwise comparable patient types.
Label / Prior	$P_{\text{train}}(Y) \neq P_{\text{test}}(Y)$	$P(\mathbf{X}\|Y)$	Disease prevalence changes, but the class-conditional measurement process remains stable.
Concept / Conditional	$P_{\text{train}}(Y\|\mathbf{X}) \neq P_{\text{test}}(Y\|\mathbf{X})$	—	Fraud strategies change; semantic drift occurs; users adapt to the deployed system.
Subpopulation	Group-conditional performance drops	—	A classifier associates ‘cow’ with ‘pasture’, failing completely on a rare subgroup while maintaining high average accuracy.

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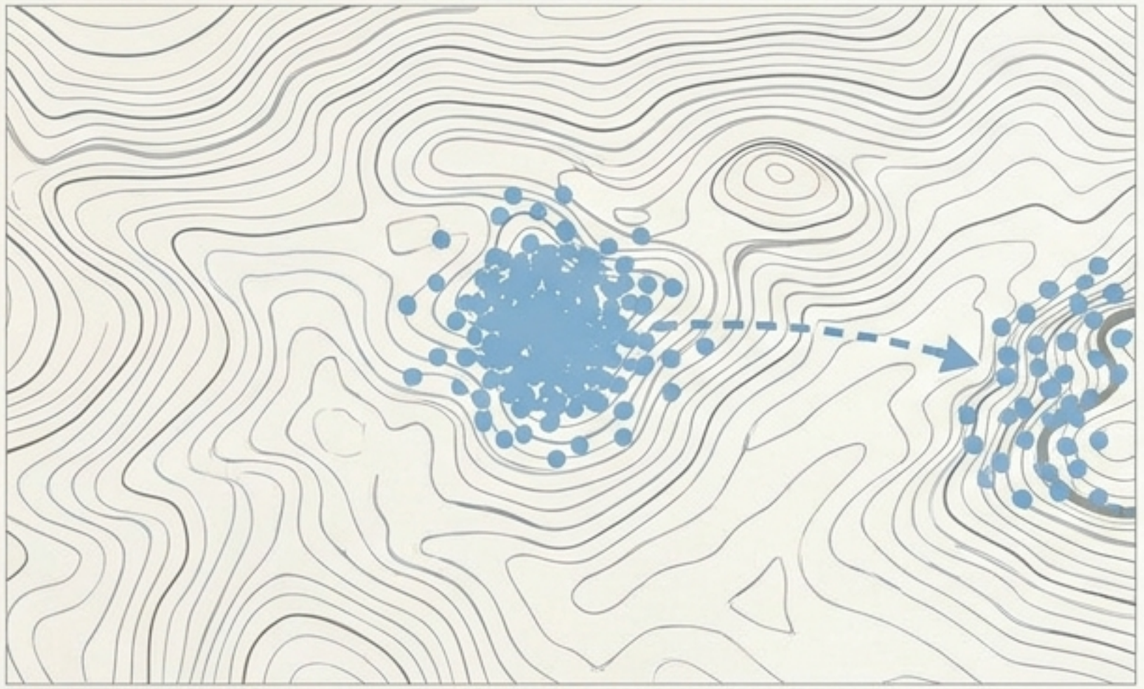
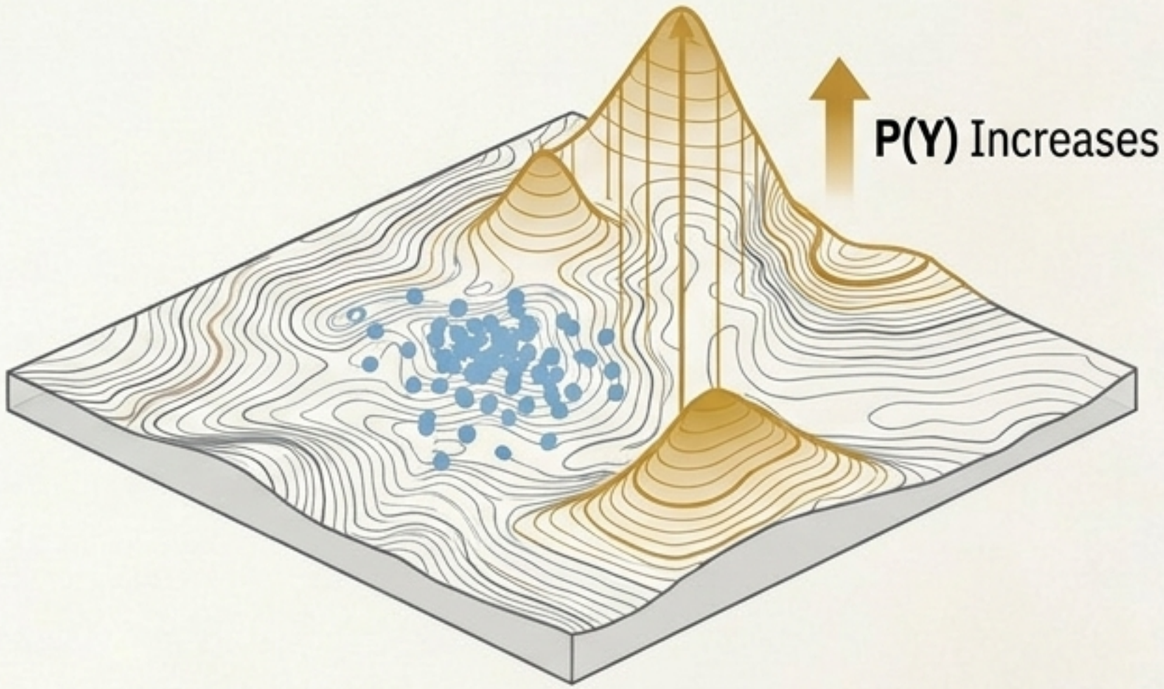
Visualizing Covariate and Label Shift

Covariate Shift: P(X) moves, **P(Y|X)** remains



Label/Prior Shift: P(Y) changes



Under Covariate Shift, importance weighting $q(\mathbf{X})/p(\mathbf{X})\ \ell(f(\mathbf{X}),\mathbf{Y})$ can mathematically express target risk, but it is operationally demanding. Large density ratios create high variance, and it fails entirely if $p(\mathbf{X})=0$ in target regions.

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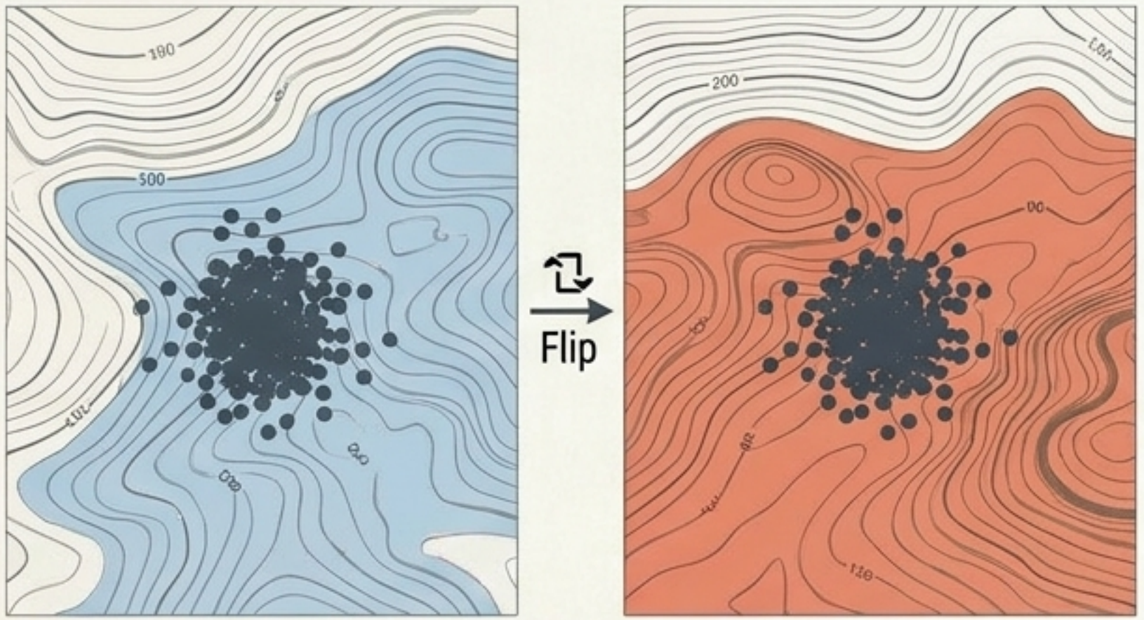
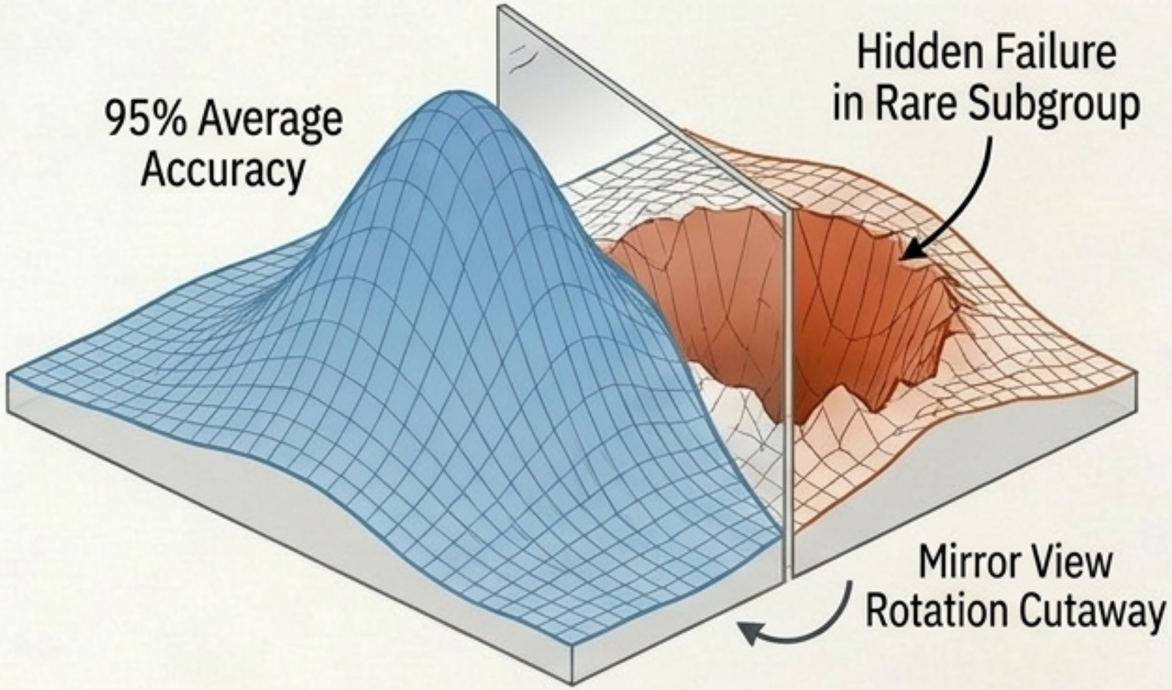
When the underlying rules of the terrain change

Concept Shift



Subpopulation Shift



Covariate reweighting alone cannot repair a changed input–outcome relationship. The definition of the target itself has drifted.

High average accuracy aggressively hides rare group failure. Evaluation must report group-conditional performance and utilize a worst-group objective $R_{\max}(f) = \max_{g \in G} R_g(f)$.

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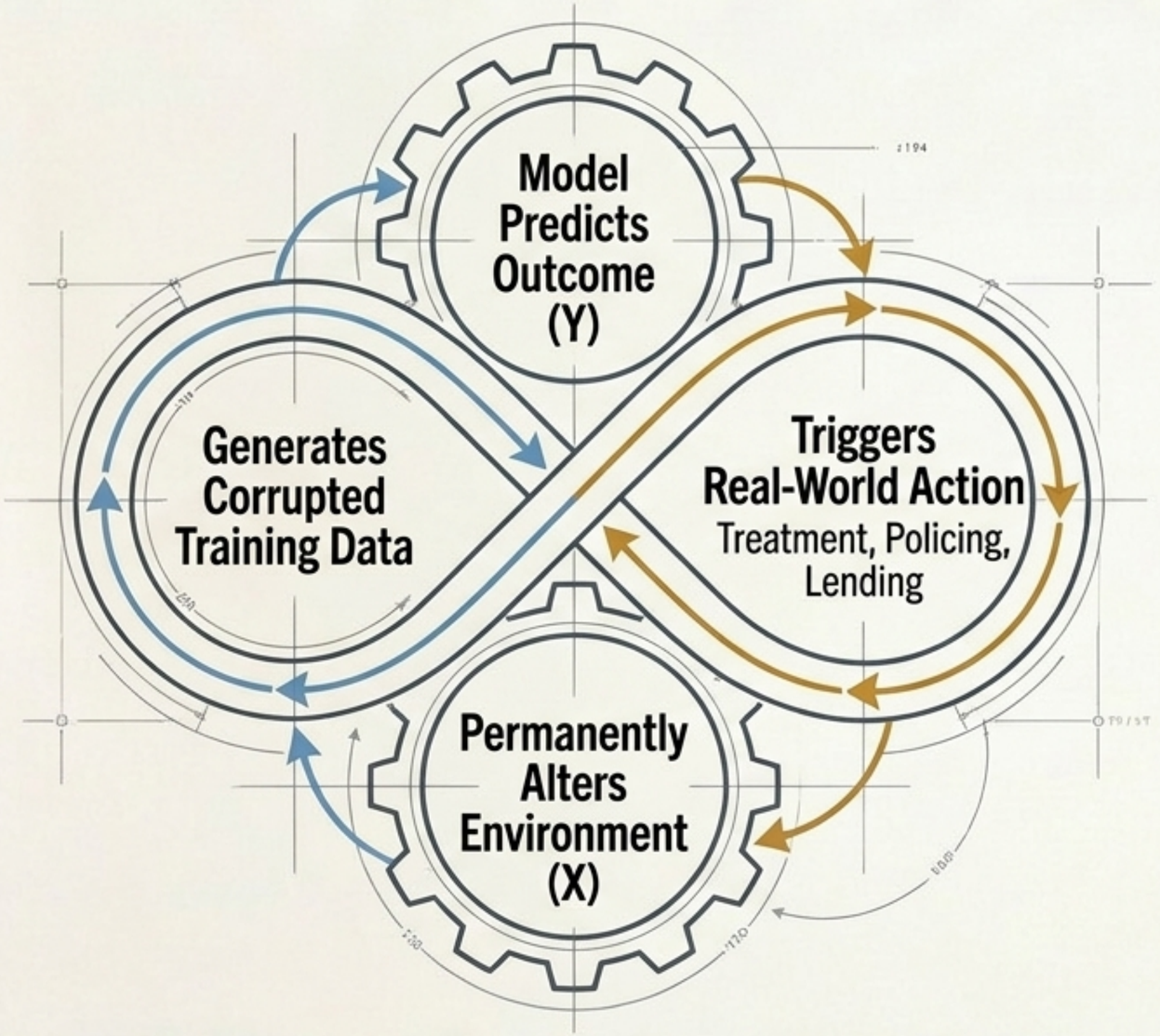
The causal feedback loop of deployment



Time, location, and policy changes produce mixtures of shifts. Feedback is the ultimate complication: a deployed policy changes which states and outcomes are observed.

A static retrospective data split cannot reproduce this causal loop.

Empirical Grounding: WILDS Benchmark

The WILDS benchmark evaluates naturally occurring shifts (e.g., hospital changes, camera locations). It proves that realistic shift matters across modalities, and that standard robustness methods do not uniformly eliminate these gaps (Koh et al.).

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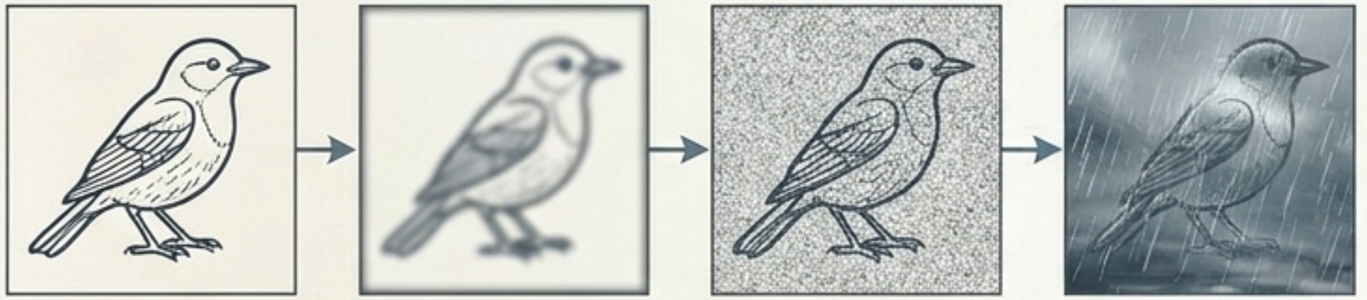
Robustness is a quantified property, not a personality trait

Natural Corruption & Benchmark Aging



Rising performance may mean pretraining data already included synthetic corruptions.

The LAION-C study showed ImageNet-C corruptions are no longer out-of-distribution for web-scale models. Proving a test is uncontaminated is nearly impossible (Li et al.).

Adversarial Vulnerability

Small, worst-case perturbations transfer across models. In Goodfellow's MNIST experiment, adversarial training reduced error from 89.4% to 17.9%, but erroneous predictions remained highly confident.

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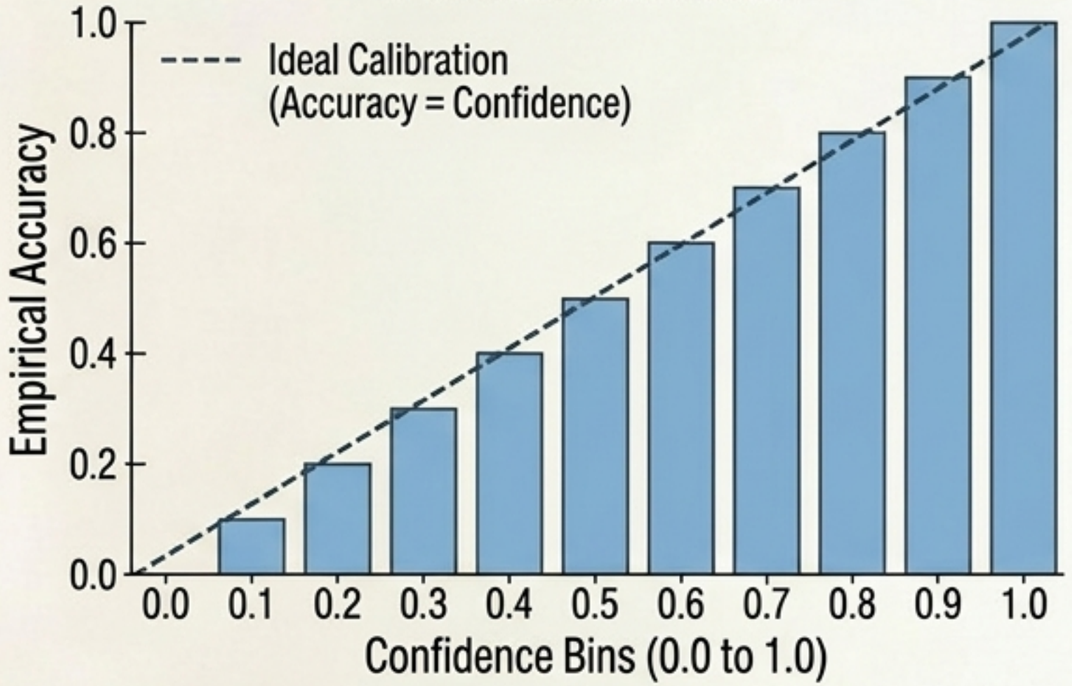
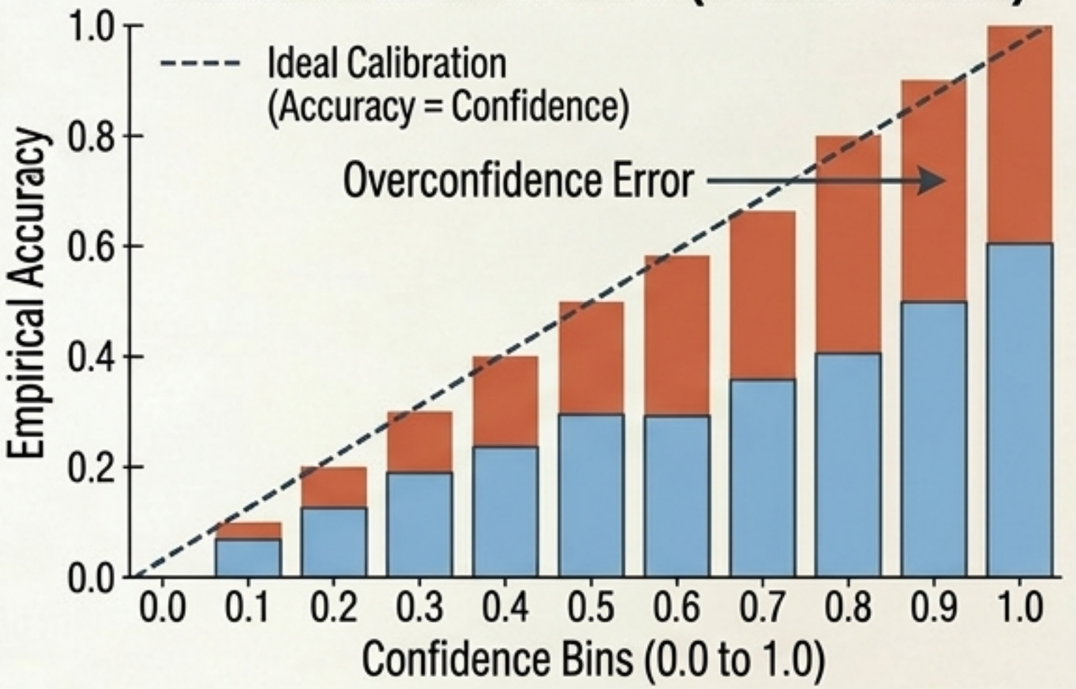
Calibration: Accuracy does not equal self-awareness

Perfect Calibration



Modern Neural Network (Overconfident)



Accuracy vs. Calibration

Accuracy measures how often a decision is correct. Calibration asks whether stated probabilities correspond to empirical frequencies. A perfectly accurate classifier can provide poor probabilities.

Key Insights & Guo et al.

Guo et al. found modern neural networks are highly overconfident. Temperature scaling preserves class ranking and improves calibration, but calibration learned on one distribution may fail entirely after shift.

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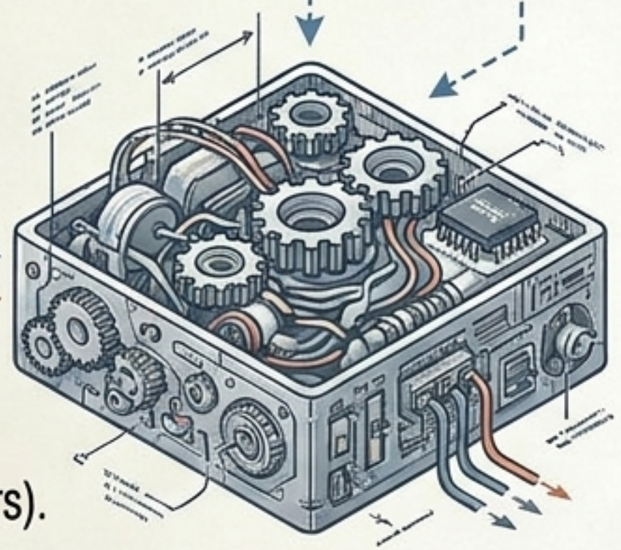
The anatomy of an aggregate score

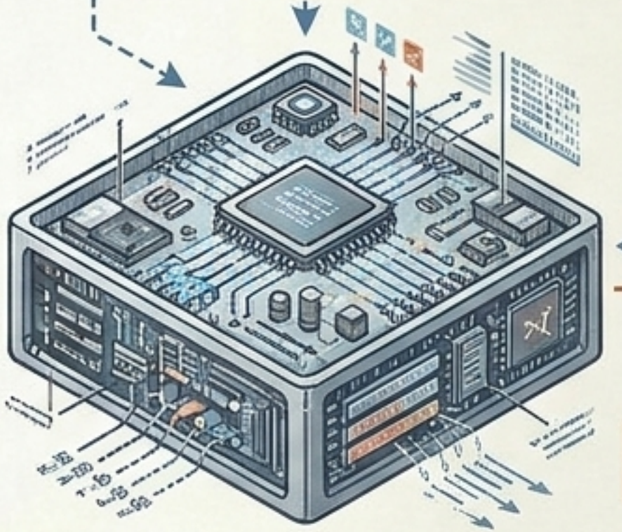
85%
Accuracy

Model A Profile

Highly Robust under natural corruption.

Severely Miscalibrated (overconfident on errors).





Model B Profile

Perfectly Calibrated probabilities.

Fails entirely on minority subpopulation.

Synthesis: Two systems with the exact same mean aggregate score can have opposite failure profiles. Evaluation is an argument supported by multi-dimensional measurements, not merely the production of a number.

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Why aggregate benchmarks actively mislead

PRECISION AUDIT CHECKLIST

- ☑ Does a micro-average allow common classes to dominate the score?
- ☑ Are missing outputs counted as failures, or ignored?
- ☑ Are subgroup sizes and confidence intervals explicitly reported?
- ☑ Was the test set used during prompt, checkpoint, or hyperparameter selection?
- ☑ Could training or retrieval data contain test questions or close paraphrases?
- ☑ Are correlated observations (same patient, same author) kept in the same split?

Statistical Note: Effect size matters. A statistically uncertain worst-group collapse demands investigation because its real-world harm is large, whereas a 0.2% mean improvement may have zero deployment value.

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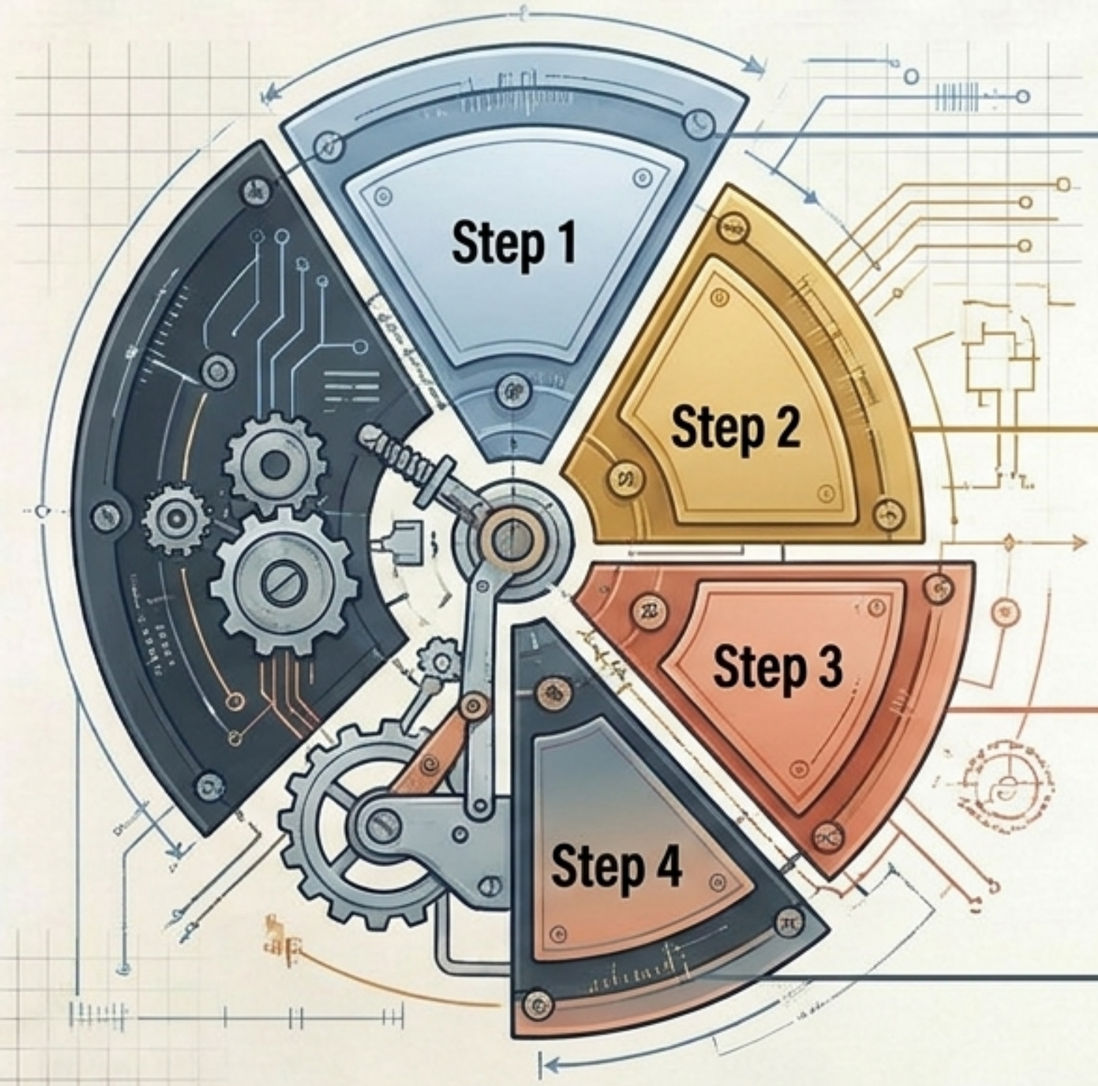
From benchmark audit to deployment evidence



Specify the claim: Define intended users, horizons, modalities, and excluded uses.
"Works on medical questions" is untestable.

Construct independent splits: Split by entities that will be new at deployment (future time, new hospital, new device). Random row-level splits are useless if observations are correlated.

Establish simple baselines: Compare against linear, rules-based, or retrieval-only systems. Retrain across random seeds to expose underspecification.

Measure multiple dimensions: Task loss, calibration, subgroup performance, abstention, latency, and consequence of failure.

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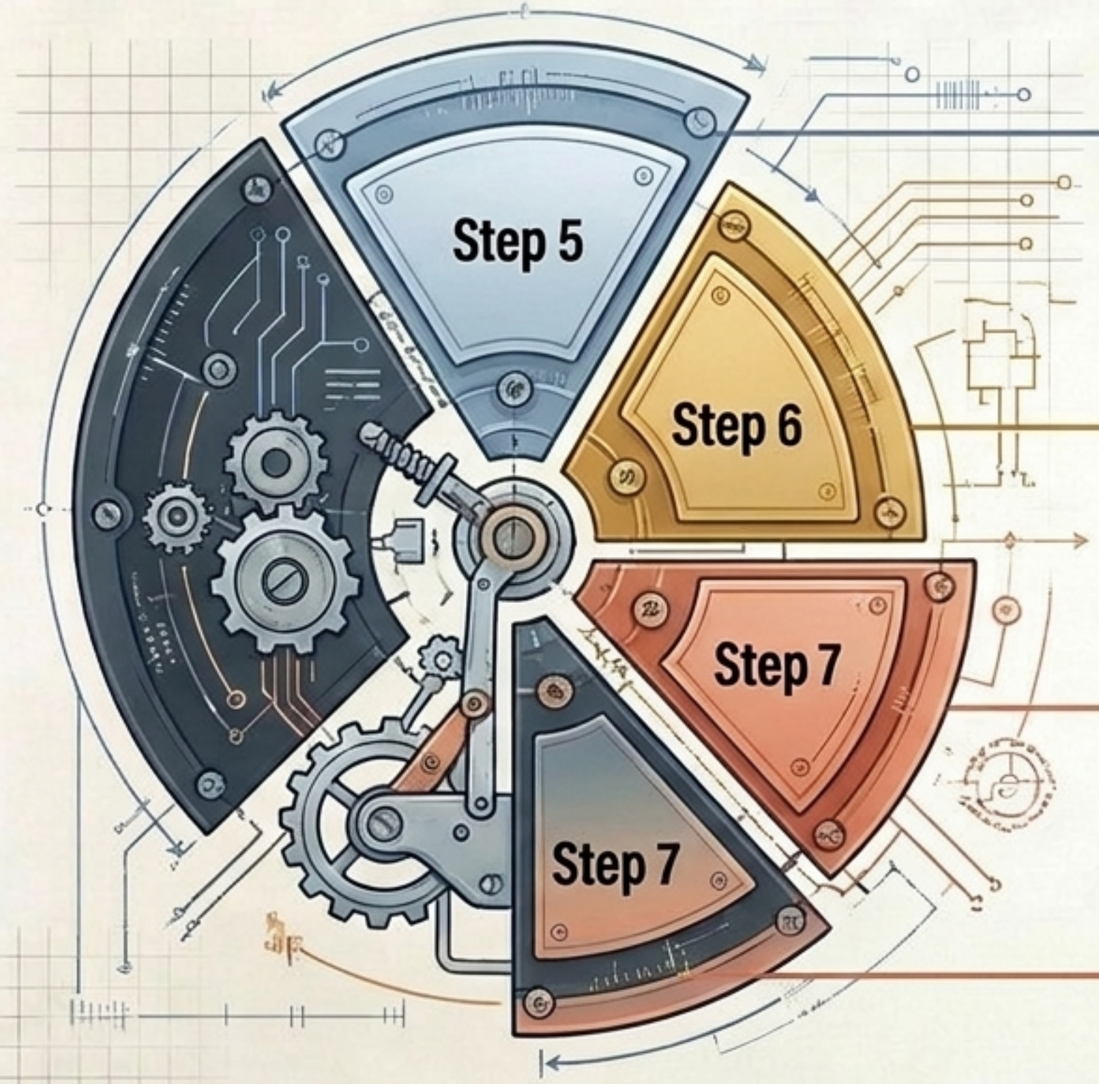
Stress testing and prospective monitoring



Stress plausible shifts: Use naturally separated target domains, corruptions, and adversarial tests. Every stress test must have causal/operational relevance.

Evaluate the whole system: Retrieval indexes, user interfaces, safety filters, and human escalation policies often dominate real performance. Foundation model evaluations must lock prompt and decoder versions.

Monitor prospectively: Input drift is an alarm, not proof of degradation. Predefine operational responses: recalibrate, route to humans, retrain, or roll back.

Monitoring without operational decision rights is just observation.

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Established facts vs. Open limitations

Established Facts

- Test performance degrades under recollected or cross-domain data, even when the task remains unchanged.
- Adversarial vulnerability, generalization, calibration, and natural-shift robustness are distinct, independent properties.
- Benchmarks inevitably lose diagnostic power as pretraining distributions absorb them.

Open Limits

- There is no complete taxonomy of future shifts.
- Proprietary pretraining data makes test contamination impossible to strictly exclude.
- Rare catastrophic failures cannot be estimated precisely from ordinary sample sizes.

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The final synthesis

The correct response is neither to abandon benchmarking nor to treat it as absolute proof. Evaluation is cumulative, scoped evidence.

*The most credible conclusion is always conditional: Under these specific data-generating **conditions**, with this **exact system configuration** and these measured limitations, **performance** met the stated **criterion**.*

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